

Mike Cummings

Mathematics and Computer 6311
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Education

University of Waterloo
PhD Combinatorics and Optimization
• Supervisor: Oliver Pechenik

Waterloo, ON
2024–2028 (expected)

McMaster University
MSc Mathematics
• Supervisor: Jenna Rajchgot
• Thesis: Gröbner Geometry for Hessenberg Varieties

Hamilton, ON
2022–2024

McMaster University
BSc Honours Mathematics and Statistics, *summa cum laude*

Hamilton, ON
2018–2022

Papers

Preprints

1. Webs and smooth components of two column Springer fibers
Submitted [arXiv:2602.16910]

Published

4. Correction to “Counting two-column Young tableaux corresponding to smooth components of Springer fibers” [J. Algebr. Comb. 61:18 (2025)] (by R. Mansour)
with Ronit Mansour
Journal of Algebraic Combinatorics **63** (2026), no. 36
3. The GeometricDecomposability package for Macaulay2
with Adam Van Tuyl
Journal of Software for Algebra and Geometry **14** (2024), no. 1, 41–50
2. Gröbner geometry for regular nilpotent Hessenberg Schubert cells
with Sergio Da Silva, Megumi Harada, and Jenna Rajchgot
Journal of Pure and Applied Algebra **228** (2024), no. 7, 107648
1. Geometric vertex decomposition and liaison for toric ideals of graphs
with Sergio Da Silva, Jenna Rajchgot, and Adam Van Tuyl
Algebraic Combinatorics **6** (2023), no. 4, 695–997

Software

1. GeometricDecomposability, a *Macaulay2* package to check whether ideals are geometrically vertex decomposable, and related methods with Adam Van Tuyl

Theses

2. Gröbner Geometry for Hessenberg Varieties
Master's thesis, McMaster University, 2024
Supervisor: Jenna Rajchgot
1. Geometric Vertex Decomposition and Hessenberg Patch Ideals
Undergraduate thesis, McMaster University, 2022
Supervisors: Sergio Da Silva, Megumi Harada, and Jenna Rajchgot

Talks

The [∨] denotes a virtual talk.

Research

Webs and smooth components of two column Springer fibers

2026 Jun. Algebraic and Enumerative Combinatorics Seminar, University of Waterloo

Gröbner geometry and multidegrees for classes of semisimple Hessenberg varieties

2025 May CanaDAM 2025, University of Ottawa

Geometric vertex decomposition: a computational technique for liaison theory

2025 Mar. Graduate Skills Seminar for the Department of Combinatorics and Optimization,
University of Waterloo

Graduate Student Combinatorics Conference, University of Southern California[∨]

Combinatorial rules for the geometry of Hessenberg varieties

2024 Nov. Algebraic and Enumerative Combinatorics Seminar, University of Waterloo

Geometric vertex decomposition: introduction and connections

2024 Feb. Graduate Student Seminar in Number Theory and Algebraic Geometry at Simon
Fraser University[∨]

Gröbner geometry for regular nilpotent Hessenberg Schubert cells

2024 Jan. Combinatorial Algebra meets Algebraic Combinatorics at LaCIM à l'Université du
Québec à Montréal

2023 Dec. Commutative Algebra session at the CMS 2023 Winter Meeting in Montréal

Mar. Algebra & Algebraic Geometry Seminar at McMaster University

Degenerations of toric ideals of graphs

2023 May Open problem session, Commutative Algebra and its Interactions with Algebraic Geometry, SLMath/CMND Summer School at the University of Notre Dame

An introduction to Hessenberg varieties

2022 Aug. Mathematics & Statistics Undergraduate Summer Research Seminar, McMaster University

Geometric vertex decomposition

2022 May Mathematics & Statistics Undergraduate Summer Research Seminar, McMaster University

Poster

Webs and smooth components of two column Springer fibers

2026 May Montreal ISM Discovery School: Combinatorics of Moduli of Curves, LACIM

Jan. CAAC 2026, Dalhousie University

Geometric vertex decomposition, substitution, and toric ideals of graphs

2025 Feb. Computational Interactions between Algebra, Combinatorics, and Discrete Geometry, IPAM

Jan. CAAC 2025 at the Fields Institute and University of Toronto

Gröbner geometry for regular nilpotent Hessenberg Schubert cells

2023 Jun. CMS 2023 Summer Meeting in Ottawa

May SLMath/CMND Summer School on Commutative Algebra and its Interactions with Algebraic Geometry at the University of Notre Dame

Expository

Webs and Springer Fibers: A case study in dynamical combinatorics and combinatorial algebraic geometry

2026 May Combinatorics and Optimization Undergraduate Research Seminar, University of Waterloo

An introduction to algebraic geometry

2024 Apr. Student Math Colloquium at Niagara University

Graph colouring, symmetric functions, and open problems

2023 Nov. Casual Math Seminar at McMaster University

Play with your food: Brussel sprouts and other games on graphs

2023 Apr. Casual Math Seminar at McMaster University

Conferences Attended

The [∨] denotes a virtual conference.

2026 Jul. CMND 2026 Thematic Program: Algebraic Combinatorics and Applications, University of Notre Dame

May Montreal ISM Discovery School: Combinatorics of Moduli of Curves, LACIM
11th Lake Michigan Workshop on Combinatorics and Graph Theory, University of Notre Dame

Mar. Cluster Algebras, Webs, and Canonical Bases, BIRS

Jan. Combinatorial Algebra meets Algebraic Combinatorics, Dalhousie University

2025 Dec. 2025 CMS Winter Meeting, Toronto

Nov. Computation in Representation Theory, ICERM

Jun. *Macaulay2* Workshop: Madison 2025, University of Wisconsin–Madison

Séminaire de Mathématiques Supérieures 2025: An Introduction to Recent Trends in Commutative Algebra, Fields Institute

May AlCoVE: an Algebraic Combinatorics Virtual Expedition 2025[∨]
CanaDAM 2025, University of Ottawa

Mar. Graduate Student Combinatorics Conference, University of Southern California[∨]

Feb. Computational Interactions between Algebra, Combinatorics, and Discrete Geometry, IPAM

Jan. Combinatorial Algebra meets Algebraic Geometry, Fields Institute and University of Toronto

2024 Jun. AlCoVE: an Algebraic Combinatorics Virtual Expedition 2024[∨]

May Macaulay2, Computational Algebraic Geometry, and String Theory, University of Utah

Why are we teaching mathematics today? FYMSiC[∨]

Jan. Combinatorial Algebra meets Algebraic Combinatorics, LaCIM à l'Université du Montréal à Québec

2023 Dec. 2023 CMS Winter Meeting, Montréal

Nov. Route 81 Conference on Commutative Algebra and Algebraic Geometry, Syracuse University

Jun. AlCoVE: an Algebraic Combinatorics Virtual Expedition 2023[∨]
2023 CMS Summer Meeting, University of Ottawa

May MSRI/SLMath CMND Summer School: Commutative Algebra and its Interactions with Algebraic Geometry, University of Notre Dame

8th Lake Michigan Workshop on Combinatorics and Graph Theory, University of Notre Dame

Teaching: The Backstage of Our Courses, FYMSiC^v

Jan. Combinatorial Algebra meets Algebraic Combinatorics, University of Waterloo

2022 Dec. 2022 CMS Winter Meeting, Toronto

Jun. ISM Discovery School: Geometry and Combinatorics of Hessenberg Varieties, Université du Québec à Montréal

Scholarships and Awards

All values Canadian dollars.

2024–27 NSERC Canada Graduate Scholarship–Doctoral, \$40,000 per year
University of Waterloo President’s Graduate Scholarship, \$10,000 per year

2024 McMaster Graduate Student Association Travel Assistance Grant, \$500

2023–24 Ontario Graduate Scholarship, \$10,000
Bernhard Banaschewski Ontario Graduate Scholarship, \$5,000
Milos Novotny Fellowship, \$4,000

2022–23 NSERC Canada Graduate Scholarship–Masters, \$17,500
Milos Novotny Fellowship, \$2,500

2022 NSERC USRA, supervisor: Adam Van Tuyl, \$8,400

2021 NSERC USRA, supervisors: Sergio Da Silva, Jenna Rajchgot, Adam Van Tuyl, \$8,120

Teaching

Instructor of Record

McMaster University

2023-24 Applications of Linear Algebra, Math 2LA3

Teaching Assistant

University of Waterloo

2025-26 Introduction to Combinatorics, Math 239
Introduction to Combinatorics (Advanced Level), Math 249
Combinatorial Algebraic Geometry, CO 439/739

2024-25 Geometry, Math 661 (Masters of Mathematics for Teachers program, CEMC)
Introduction to Optimization, CO 250
Introduction to Combinatorics (Non-Specialist Level), Math 229

McMaster University

- 2023-24 Theory of Linear Algebra, Math 2R03
Abstract Algebra, Math 3GR3
- 2022-23 Theory of Linear Algebra, Math 2R03
Calculus for Math and Stats I, Math 1X03
- 2021-22 Calculus for Math and Stats II, Math 1XX3
Calculus for Math and Stats I, Math 1X03
- 2020-21 Introduction to Mathematical Scientific Computation, Math 1MP3
Matlab for Linear Algebra I, Math 1B03/1ZC3
Linear Algebra I (Engineering Math II-B), Math 1ZC3
Calculus for Math and Stats I, Math 1X03
- 2019-20 Introduction to Mathematical Scientific Computation, Math 1MP3

Professional Activities

Service

- 2025– Graduate Student Representative for the Department of Combinatorics and Optimization, University of Waterloo

Organization

- 2023–24 McMaster Casual Math Seminar
with Alexi Block Gorman (Winter 2023), Justin Gebel (Fall 2023–Winter 2024), Dicle Mutlu, and Adele Padgett
- 2022 McMaster Math & Stats Undergraduate Summer Research Seminar
with Hainan Xu

Referee

- (1) Combinatorial Commutative Algebra
- (1) Electronic Journal of Combinatorics
- (1) Proceedings of the American Mathematical Society

Professional Membership

- 2022– Canadian Mathematical Society
American Mathematical Society